

Project Euler Problem 2: Even Fibonacci Numbers

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Problem

Each new term in the Fibonacci sequence is generated by adding the previous two terms. By starting with 1 and 2, the first 10 terms will be:

$$1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Proof

Definition 1. The n th Fibonacci number $F(n)$ is given by the following relation:

$$F(n) = F(n-1) + F(n-2) \quad \text{for } n > 2$$

$$F(1) = 1$$

$$F(2) = 2$$

One interesting property that arises from this definition is that starting from $F(2)$, every third Fibonacci number is even.

Lemma 2. For $n \geq 0$, $F(3n+1)$ is odd, $F(3n+2)$ is even, and $F(3n+3)$ is odd.

Proof. We proceed by induction on n .

Base case ($n = 0$). Definition 1 gives this almost immediately: $F(1) = 1$ is odd, $F(2) = 2$ is even, and

$$F(3) = F(1) + F(2) = 3$$

is odd. Thus, all three statements hold for $n = 0$.

Inductive step. Suppose that for some $n \geq 0$, all three statements hold. We show they also hold for $n + 1$. First,

$$F(3n+4) = F(3n+3) + F(3n+2)$$

We know that $F(3n+3)$ is odd and $F(3n+2)$ is even. Thus, $F(3n+4)$ must be odd, since the sum of an odd and an even number is odd.

Next,

$$F(3n+5) = F(3n+4) + F(3n+3)$$

We just showed that $F(3n+4)$ is odd, and by hypothesis $F(3n+3)$ is odd. Thus, $F(3n+5)$ must be even, since the sum of two odd numbers is even.

Finally,

$$F(3n+6) = F(3n+5) + F(3n+4)$$

We have shown that $F(3n+5)$ is even and $F(3n+4)$ is odd. Thus, $F(3n+6)$ must be odd, since the sum of an even and an odd number is odd.

Since $3(n+1)+1 = 3n+4$, $3(n+1)+2 = 3n+5$, and $3(n+1)+3 = 3n+6$, all three statements hold for $n+1$. By induction, they hold for all $n \geq 0$. $\square$

Corollary 3. *For $n \geq 1$, $F(n)$ is even if and only if $n \equiv 2 \pmod{3}$.*

Proof. Every $n \geq 1$ can be written in exactly one of the forms $3k+1$, $3k+2$, or $3k+3$ for some $k \geq 0$. By lemma 2, $F(n)$ is even in the second case and odd in the other two. Thus, $F(n)$ is even exactly when $n = 3k+2$, that is, when $n \equiv 2 \pmod{3}$. $\square$

Following Corollary 3, we can develop a recurrence relation that generates the even Fibonacci numbers directly, skipping the odd ones.

Lemma 4. *Let $E(n)$ be the n th even Fibonacci number.*

$$\begin{aligned} E(n) &= 4E(n-1) + E(n-2) && \text{for } n > 2 \\ E(1) &= 2 \\ E(2) &= 8 \end{aligned}$$

Proof. By Corollary 3, the even terms are exactly $F(2), F(5), F(8), \dots$, so $E(n) = F(3n-1)$. This gives the base cases directly: $E(1) = F(2) = 2$ and $E(2) = F(5) = 8$.

Now let $n > 2$ and $m = 3n-4$. Then $E(n) = F(m+3)$, $E(n-1) = F(m)$, and $E(n-2) = F(m-3)$.

With this formulation, it suffices to show that $F(m+3) = 4F(m) + F(m-3)$.

$$\begin{aligned} F(m+3) &= F(m+2) + F(m+1) && \text{by definition} \\ &= 2F(m+1) + F(m) && \text{expanding } F(m+2) \\ &= 3F(m) + 2F(m-1) && \text{expanding } F(m+1) \\ &= 3F(m) + F(m-1) + F(m-2) && \text{expanding one copy of } F(m-1) \\ &\quad + F(m-3) \\ &= 4F(m) + F(m-3) && \text{regrouping by Definition 1} \end{aligned}$$

Substituting back gives $E(n) = 4E(n-1) + E(n-2)$, as required. $\square$

Theorem 5. *The sum of the even Fibonacci numbers not exceeding 4,000,000 is 4,613,732.*

Proof. By Corollary 3, the even Fibonacci numbers are exactly the terms $E(k)$. The sequence E is strictly increasing: $E(2) > E(1) > 0$, and for $k > 2$, $E(k) = 4E(k-1) + E(k-2) > E(k-1)$. So we can generate the terms in order with Lemma 4 and stop at the first one that exceeds 4,000,000.

k	$E(k)$	$\sum_{i=1}^k E(i)$
1	2	2
2	8	10
3	34	44
4	144	188
5	610	798
6	2,584	3,382
7	10,946	14,328
8	46,368	60,696
9	196,418	257,114
10	832,040	1,089,154
11	3,524,578	4,613,732
12	14,930,352	—

Since $E(12) > 4,000,000$, the even terms not exceeding 4,000,000 are $E(1)$ through $E(11)$, and their sum is 4,613,732. $\square$

Generating only the even terms takes 11 steps, whereas the 32 Fibonacci terms up to 4,000,000 would each need to be generated and checked for parity.